

Reconsidering the p-value

Readings for today

- Wasserstein, R. L., Schirm, A. L., & Lazar, N. A. (2019). Moving to a World Beyond " $p < 0.05$ ". *The American Statistician*, 73(S1), 1-19.

Topics

1. Null hypothesis test statistics (NHTSs)
2. ATOM approach

Null hypothesis test statistics

The replication crisis

Problem: Uncontrolled sources of variability can lead to dramatically different findings, even when using identical methods.

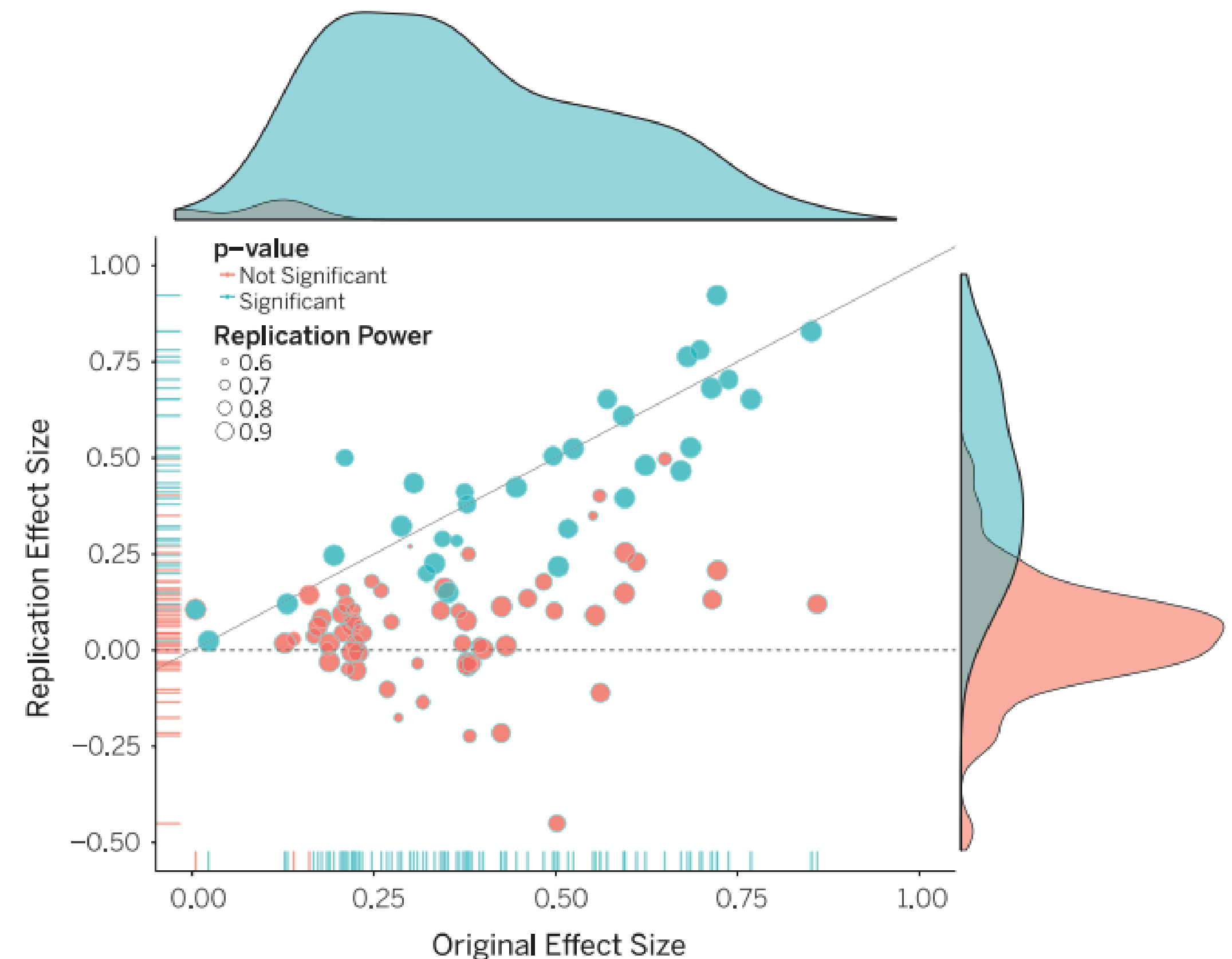
RESEARCH ARTICLE SUMMARY

PSYCHOLOGY

Estimating the reproducibility of psychological science

Open Science Collaboration*

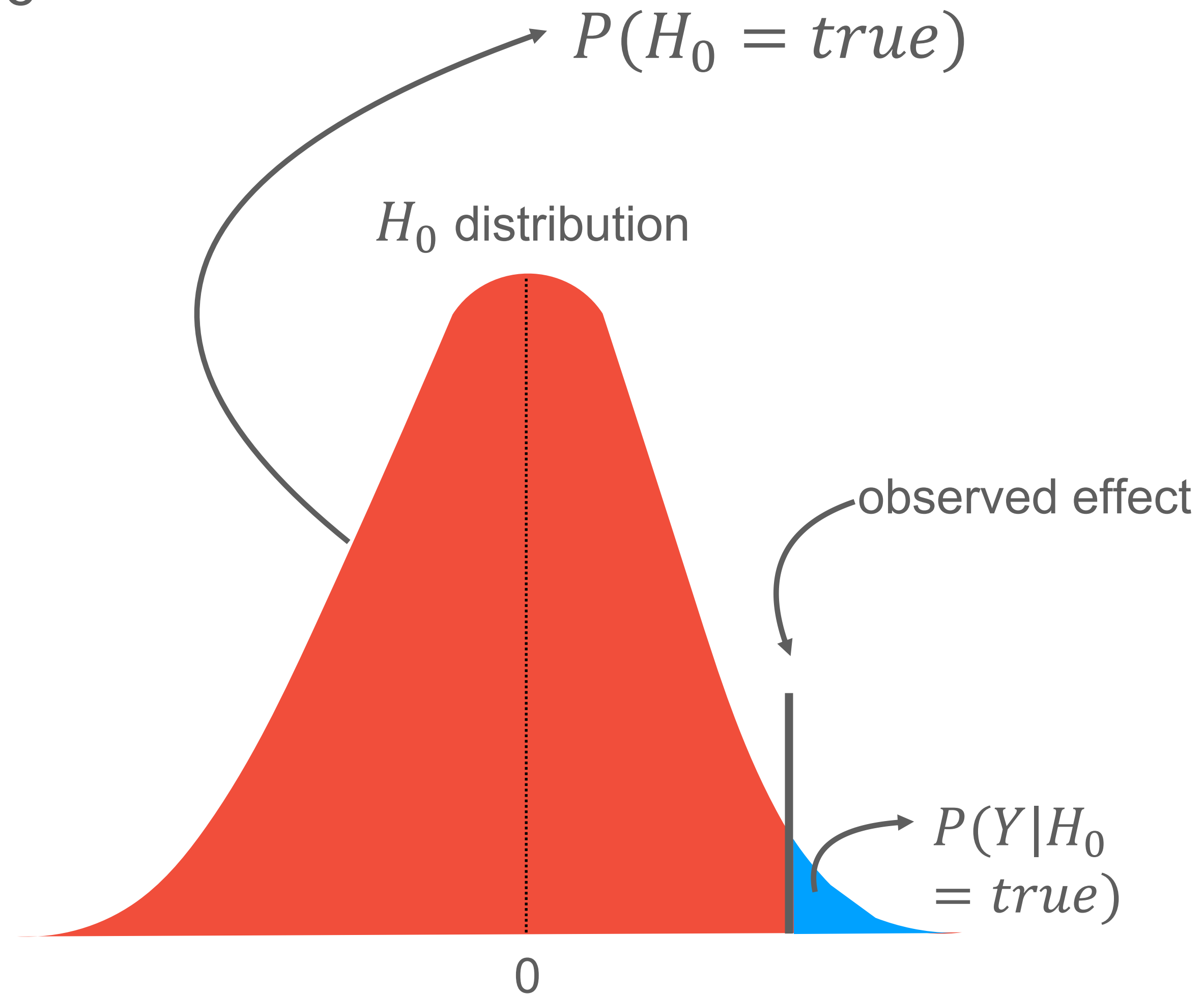
(Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251).



The null hypothesis (H_0)

H_0 : The probability that you observe your data if your predicted relationships are not true.

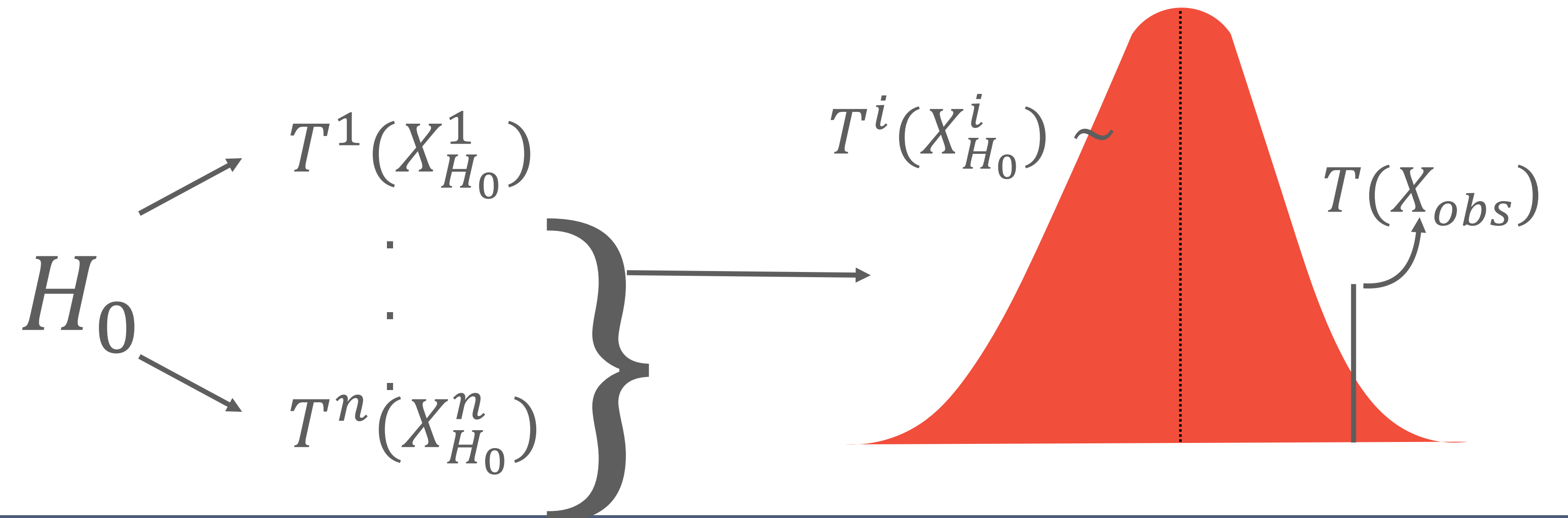
Sometimes “chance”, but not always.



Null hypothesis test statistics (NHTSs)

$$p = P(Y|H_0 = \text{true}) = P(T(X_{H_0}) \geq T(X_{obs}))$$

observed data null hypothesis test statistic predictor variables generated by H_0 predictor variables you observe



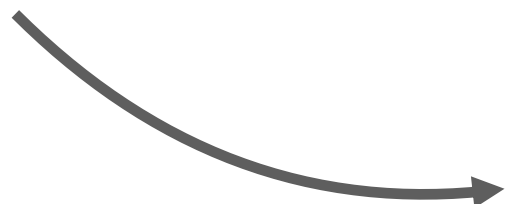
Problems with NHTSs

Fundamental limitations:

1. Depend on unobserved data. ← true even for permutation tests.
2. Depend on subjective intentions. ← e.g., p-hacking
3. Do not quantify statistical evidence. ← existence $\neq$ evidence
4. Poorly understood. ← e.g., $p=0.051$ is not that different than $p=0.049$.

Poor practices with NHTSs

NHTSs (e.g., p-values) are simply meant to be a flag to indicate whether results warrant further inspection.



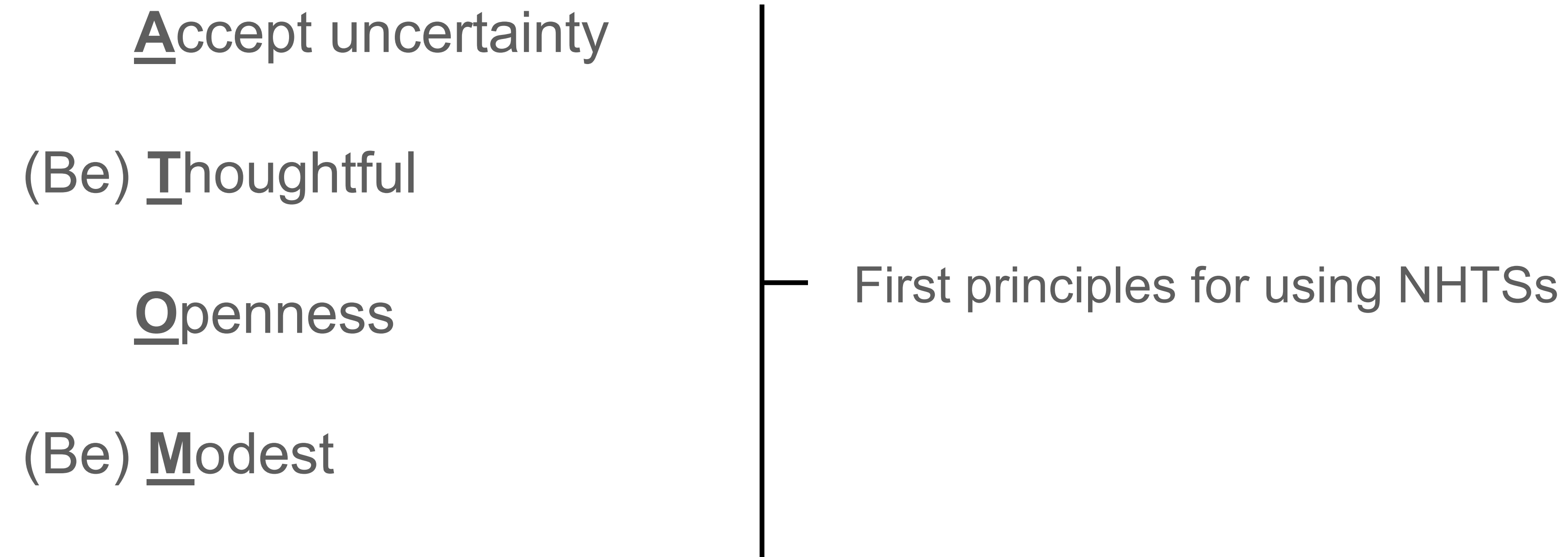
p-values are the beginning, not the end.

Common poor practices:

- Concluding solely on “statistical significance” (e.g., $p < 0.05$)
- Believing that an effect exists simply because of “statistical significance”
- Believing that an effect does not exist because of lack of “statistical significance”
- Believing that the p-value gives you the probability of “chance” alone.

ATOM approach

ATOM



Accept uncertainty

Acknowledge that uncertainty exists & test for it.

1. Seek better or more comprehensive measures of uncertainty. ← always measure uncertainty
2. Increase focus on data quality and experimental design. ← garbage in, garbage out
3. Move away from the false certainty of “statistical significance” ← p-value $\neq$ truth

(Be) Thoughtful

Develop a holistic approach to inferring from your data.

 look at the whole story!

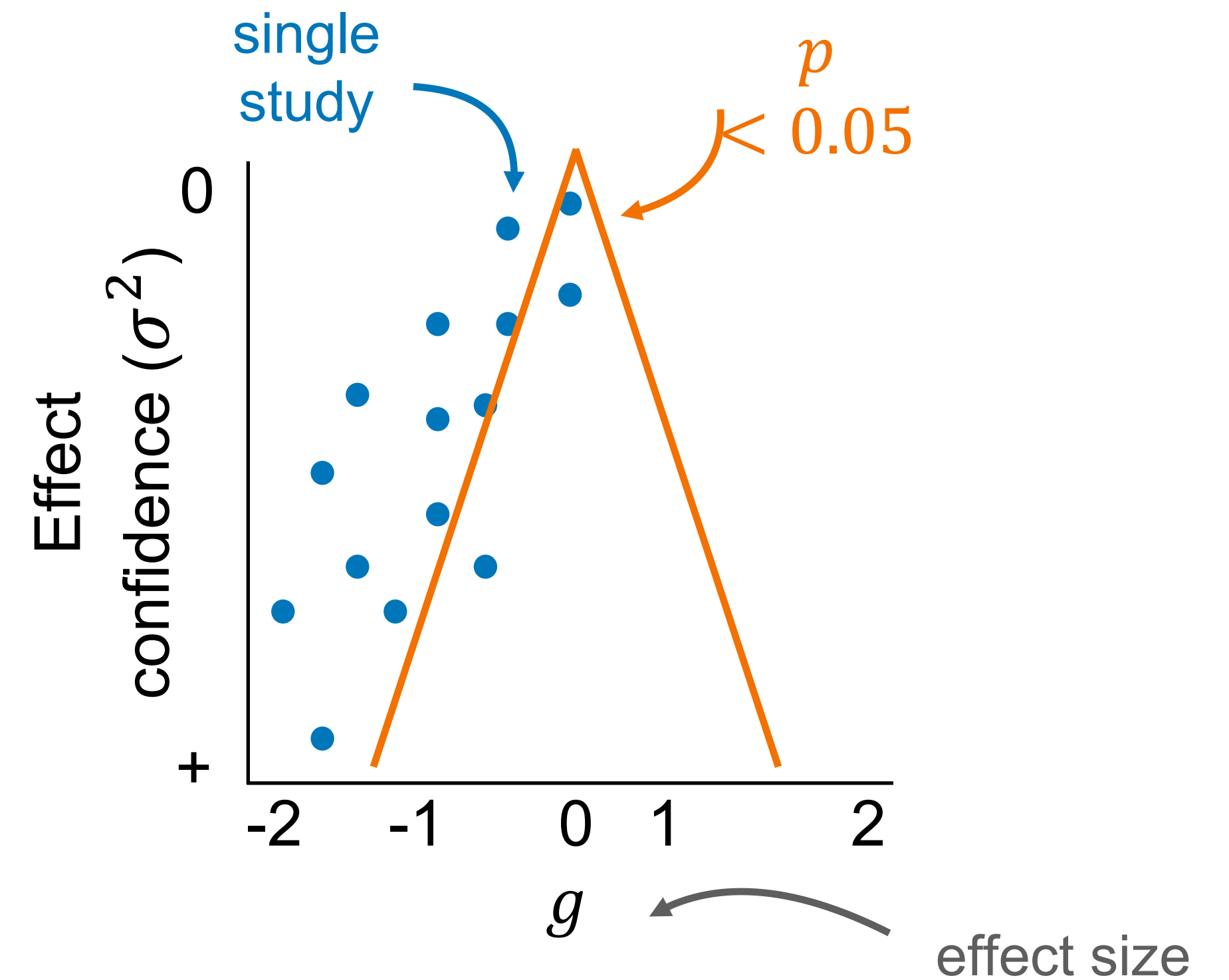
1. Be mindful of data quality at all stages.
2. Consider context & prior evidence.
3. Look ahead to how results predict future outcomes.
4. Be mindful of effect sizes.
5. Consider both mechanisms and possible confounds.
6. Think of your entire toolbox of methods.
7. Effectively communicate your uncertainty in different ways.

Openness

Adopt “open science” practices

1. Release your code.
2. Release your data.
3. Communicate your methods clearly & completely.
4. Maximize replicability.

Signature of “file drawer” problem



selection bias = $\uparrow \sigma^2$ with $\uparrow g$

(Be) Modest

Being overzealous and inflating conclusions is only effective in the short-term.

1. There is no “perfect study”.
2. Inflation of conclusions reduces replicability.
3. Be a neutral judge of your findings.
 - Temper conclusions
 - Encourage others to replicate

Aggrandizement is a means of impairing confidence in your findings.

ATOM

Accept uncertainty

(Be) Thoughtful

Openness

(Be) Modest

Think of NHSTs as a single tool
in an arsenal of many tools that
look at your data in different
ways.

Take home message

- The p-value is a very useful existential test of your null hypothesis, but should be used in conjunction with many other tests to draw reliable conclusions from your data.